

SUPPLEMENT TO “NEURAL BELLMAN OPERATORS”

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This supplement records the complete prospective-service registry, numerical semantics, confidence intervals, datewise error accounts and source-preservation map for Neural Bellman Operators. The active mathematical statements are proved in the article; all preceding proofs and applications remain in the complete companions.

1. EXECUTION, NUMERICAL SEMANTICS AND COMPLETE RECORDS

The active article retains the constructive own-future NBO operator and adds trained neural action search, non-tensor verification, prospective economic stopping and validation-qualified learned structure. Its mathematical statements are proved in the article. This supplement records the execution contract and all service outcomes. The complete development and proof companions retain every preceding theorem, application and adverse experiment under its original assumptions.

1.1. *Source and outcome identity*

The controlling advisory report reviews manuscript commit 00e837adb431f4d4b5248fc6d5ff1fc927dd3b65. Its review commit is adf1256cff9cde365246a3db2dac90c72fda3b13. The separately frozen primary and signed-sensitivity executions are inherited from 7f4de488134e93b6cfb5dd95a3e6d7e14b45ac2a. This manuscript integration does not rerun those production services or add cost samples. The source freezes, interrupted controller, corrected controller, fitted parameters, raw endpoints, clocks and immutable process checkpoints remain unchanged.

The publication audit reloads each saved fitted critic, reconstructs its proposal, and regenerates each unique whole-domain certificate with the frozen numerical algorithms. Every saved candidate endpoint, covering endpoint, residual band, value band and policy choice is compared with that calculation. Repetition-equivalent inputs are computed once and reused only after their canonical input identities match. This is primitive re-evaluation of the recorded certificate with the same

algorithm, not an independent proof assistant or an independent implementation of the entire economic simulator.

All 132 services are then checked for their exact incumbent transfer, nine common candidates and tenth proposal, complete continuous-action cover, strict minimum-upper selection, feasibility, exact finite-sweep recurrence, declared look order, cumulative sample prefixes and first legal stopping decision. Every referenced file and process-log hash is checked. No later inference or construction is permitted after a target crossing. A finite budget exhaustion remains in the outcome table.

1.2. *Cross-platform moment verification*

Different numerical-library versions can differ in the final rounding of a sample second moment even when their endpoint arrays are byte identical. The audit therefore does not overwrite a scientific interval with a newly rounded interval and call it the same record. Instead, it converts each binary64 endpoint to an exact rational, computes its exact sample mean, second moment and unbiased variance, and checks their containment in the originally stored outward moment account. It reconstructs the rational empirical-Bernstein endpoints from those stored accounts. Thus a valid original interval can be verified without assuming bit identity of a library reduction on another platform.

Validation confidence boxes receive the corresponding exact-mean check, with their stored confidence radius and numerical-bin allowance. All policy-cost endpoints retain signed path gains. Conditional expected-cost safety does not imply that a common-shock pathwise gain is nonnegative; such paths are not clipped away. The declared independent-bin statistical model remains distinct from the reproducibility of a pseudorandom seed.

1.3. *Residual cache and signed tubes*

A range query includes every closed leaf meeting the query box. This is conservative at a boundary but preserves the half-open deployed observation rule. No smoothness is imposed on the actor. The first-stage installed policy is globally zero. Sensitivity is used only under that verified reference identity; a changed incumbent

falls back to the generic cache. The forward zero-policy tube contains the entire future innovation support and the backward tube uses the sparse Jacobian of the original drift. Its first innovation is integrated over the declared bins with exact probabilities. The algorithm does not recursively enumerate all q^T future bins.

Each raw candidate and covering comparison retains its base-cache interval, signed-tube interval and intersection. Empty intersections are errors, not opportunities to choose a midpoint. Table entries for $U - C$ and $C - L$ are maxima of independently nonnegative cellwise quantities. Their separate maxima need not sum to the displayed maximum of $U - L$. Refinement attribution is a comparison of valid enclosures, not a causal decomposition of state discretization, action search, stochastic integration and floating-point error.

1.4. *Exact neural action search*

The new scalar-action solver reads the exact stored dyadic coefficients. It enumerates rational hinge breakpoints, constructs the cubic derivative of each quartic piece, isolates every real stationary root by rational intervals, and compares candidate lattice actions by exact rational objective values. Multiple roots and zero innovation projections have explicit tests. The retained saved-model regressions specify one acquired-cell center in each task before checking its entire finite lattice exhaustively. They create neither a replacement production policy nor a new economic observation. Root-isolation cost and coefficient bit lengths are not replaced by a unit-cost count in a runtime claim.

2. COMPLETE PROSPECTIVE-SERVICE REGISTRY

TABLE I: Every prospective service and timing repetition

Cohort	d	T	Method	q	Rep.	Stages	Paths	End	Work (s)
P 2 2	W	49/50	0	1	16384	H	1.149		
P 2 2	W	49/50	1	1	16384	H	1.144		
P 2 2	W	49/50	2	1	16384	H	1.153		
P 2 2	W	9/10	0	2	65536	H	4.664		

	Cohort	d	T	Method	q	Rep.	Stages	Paths	End	Work (s)	
1											1
2											2
3		P 2	2	W	9/10	1	2	65536	H	4.617	3
4		P 2	2	W	9/10	2	2	65536	H	4.621	4
5		P 2	2	E	49/50	0	1	16384	H	2.719	5
6		P 2	2	E	49/50	1	1	16384	H	2.714	6
7		P 2	2	E	49/50	2	1	16384	H	2.723	7
8		P 2	2	E	9/10	0	2	65536	H	10.442	8
9		P 2	2	E	9/10	1	2	65536	H	10.313	9
10		P 2	2	E	9/10	2	2	65536	H	10.247	10
11		P 4	4	E	49/50	0	2	16384	H	37.304	11
12		P 4	4	E	49/50	1	2	16384	H	36.875	12
13		P 4	4	E	49/50	2	2	16384	H	37.219	13
14		P 4	4	E	9/10	0	2	16384	E	35.003	14
15		P 4	4	E	9/10	1	2	16384	E	35.275	15
16		P 4	4	E	9/10	2	2	16384	E	35.054	16
17		P 8	6	E	49/50	0	2	65536	E	47.362	17
18		P 8	6	E	49/50	1	2	65536	E	48.146	18
19		P 8	6	E	49/50	2	2	65536	E	47.742	19
20		P 8	6	E	9/10	0	2	4096	E	43.171	20
21		P 8	6	E	9/10	1	2	4096	E	44.311	21
22		P 8	6	E	9/10	2	2	4096	E	43.425	22
23		P 2	2	Q	49/50	0	1	16384	H	0.936	23
24		P 2	2	Q	49/50	1	1	16384	H	0.933	24
25		P 2	2	Q	49/50	2	1	16384	H	0.928	25
26		P 2	2	Q	9/10	0	2	65536	H	3.900	26
27		P 2	2	Q	9/10	1	2	65536	H	3.904	27
28		P 2	2	Q	9/10	2	2	65536	H	3.879	28
29		P 4	4	Q	49/50	0	2	16384	H	10.739	29
30		P 4	4	Q	49/50	1	2	16384	H	10.567	30
31		P 4	4	Q	49/50	2	2	16384	H	10.715	31
32		P 4	4	Q	9/10	0	2	16384	E	9.153	32
		P 4	4	Q	9/10	1	2	16384	E	8.714	
		P 4	4	Q	9/10	2	2	16384	E	8.462	
		P 8	6	Q	49/50	0	2	65536	E	17.204	

	Cohort	d	T	Method	q	Rep.	Stages	Paths	End	Work (s)	
1											1
2											2
3	P	8	6	Q	49/50	1	2	65536	E	16.978	3
4	P	8	6	Q	49/50	2	2	65536	E	17.123	4
5	P	8	6	Q	9/10	0	2	4096	E	13.132	5
6	P	8	6	Q	9/10	1	2	4096	E	13.187	6
7	P	8	6	Q	9/10	2	2	4096	E	13.249	7
8	P	2	2	N	49/50	0	1	16384	H	2.372	8
9	P	2	2	N	49/50	1	1	16384	H	2.376	9
10	P	2	2	N	49/50	2	1	16384	H	2.381	10
11	P	2	2	N	9/10	0	2	65536	H	7.159	11
12	P	2	2	N	9/10	1	2	65536	H	7.114	12
13	P	2	2	N	9/10	2	2	65536	H	7.112	13
14	P	4	4	N	49/50	0	2	16384	H	18.806	14
15	P	4	4	N	49/50	1	2	16384	H	18.506	15
16	P	4	4	N	49/50	2	2	16384	H	18.913	16
17	P	4	4	N	9/10	0	2	16384	E	17.163	17
18	P	4	4	N	9/10	1	2	16384	E	17.608	18
19	P	4	4	N	9/10	2	2	16384	E	17.196	19
20	P	8	6	N	49/50	0	2	65536	E	23.934	20
21	P	8	6	N	49/50	1	2	65536	E	24.386	21
22	P	8	6	N	49/50	2	2	65536	E	23.957	22
23	P	8	6	N	9/10	0	2	4096	E	19.988	23
24	P	8	6	N	9/10	1	2	4096	E	20.054	24
25	P	8	6	N	9/10	2	2	4096	E	19.783	25
26	P	2	2	F	49/50	0	1	16384	H	1.202	26
27	P	2	2	F	49/50	1	1	16384	H	1.195	27
28	P	2	2	F	49/50	2	1	16384	H	1.195	28
29	P	2	2	F	9/10	0	2	65536	H	4.761	29
30	P	2	2	F	9/10	1	2	65536	H	4.729	30
31	P	2	2	F	9/10	2	2	65536	H	4.708	31
32	S	2	2	W	49/50	0	1	4096	H	0.727	32
	S	2	2	W	49/50	1	1	4096	H	0.739	
	S	2	2	W	49/50	2	1	4096	H	0.888	
	S	2	2	W	9/10	0	1	16384	H	0.808	

	Cohort	d	T	Method	q	Rep.	Stages	Paths	End	Work (s)		
1											1	
2											2	
3		S	2	2	W	9/10	1	1	16384	H	0.781	3
4		S	2	2	W	9/10	2	1	16384	H	1.345	4
5		S	2	2	E	49/50	0	1	4096	H	1.620	5
6		S	2	2	E	49/50	1	1	4096	H	1.865	6
7		S	2	2	E	49/50	2	1	4096	H	1.688	7
8		S	2	2	E	9/10	0	1	16384	H	1.740	8
9		S	2	2	E	9/10	1	1	16384	H	1.678	9
10		S	2	2	E	9/10	2	1	16384	H	1.737	10
11		S	4	4	E	49/50	0	1	4096	H	11.605	11
12		S	4	4	E	49/50	1	1	4096	H	11.700	12
13		S	4	4	E	49/50	2	1	4096	H	11.589	13
14		S	4	4	E	9/10	0	1	4096	H	11.781	14
15		S	4	4	E	9/10	1	1	4096	H	11.515	15
16		S	4	4	E	9/10	2	1	4096	H	11.684	16
17		S	8	6	E	49/50	0	1	4096	H	31.428	17
18		S	8	6	E	49/50	1	1	4096	H	31.032	18
19		S	8	6	E	49/50	2	1	4096	H	31.304	19
20		S	8	6	E	9/10	0	1	16384	H	31.833	20
21		S	8	6	E	9/10	1	1	16384	H	31.886	21
22		S	8	6	E	9/10	2	1	16384	H	35.303	22
23		S	2	2	Q	49/50	0	1	4096	H	0.619	23
24		S	2	2	Q	49/50	1	1	4096	H	0.665	24
25		S	2	2	Q	49/50	2	1	4096	H	0.660	25
26		S	2	2	Q	9/10	0	1	16384	H	0.695	26
27		S	2	2	Q	9/10	1	1	16384	H	0.701	27
28		S	2	2	Q	9/10	2	1	16384	H	0.745	28
29		S	4	4	Q	49/50	0	1	4096	H	5.539	29
30		S	4	4	Q	49/50	1	1	4096	H	5.559	30
31		S	4	4	Q	49/50	2	1	4096	H	5.619	31
32		S	4	4	Q	9/10	0	1	4096	H	5.582	32
		S	4	4	Q	9/10	1	1	4096	H	5.576	
		S	4	4	Q	9/10	2	1	4096	H	5.583	
		S	8	6	Q	49/50	0	1	4096	H	20.344	

	Cohort	d	T	Method	q	Rep.	Stages	Paths	End	Work (s)	
1											1
2											2
3	S	8	6	Q	49/50	1	1	4096	H	20.383	3
4	S	8	6	Q	49/50	2	1	4096	H	21.123	4
5	S	8	6	Q	9/10	0	1	16384	H	21.243	5
6	S	8	6	Q	9/10	1	1	16384	H	21.027	6
7	S	8	6	Q	9/10	2	1	16384	H	21.882	7
8	S	2	2	N	49/50	0	1	4096	H	1.304	8
9	S	2	2	N	49/50	1	1	4096	H	1.333	9
10	S	2	2	N	49/50	2	1	4096	H	1.292	10
11	S	2	2	N	9/10	0	1	16384	H	1.431	11
12	S	2	2	N	9/10	1	1	16384	H	1.389	12
13	S	2	2	N	9/10	2	1	16384	H	1.409	13
14	S	4	4	N	49/50	0	1	4096	H	9.111	14
15	S	4	4	N	49/50	1	1	4096	H	9.106	15
16	S	4	4	N	49/50	2	1	4096	H	9.140	16
17	S	4	4	N	9/10	0	1	4096	H	9.238	17
18	S	4	4	N	9/10	1	1	4096	H	9.087	18
19	S	4	4	N	9/10	2	1	4096	H	9.073	19
20	S	8	6	N	49/50	0	1	4096	H	25.945	20
21	S	8	6	N	49/50	1	1	4096	H	26.944	21
22	S	8	6	N	49/50	2	1	4096	H	25.865	22
23	S	8	6	N	9/10	0	1	16384	H	26.714	23
24	S	8	6	N	9/10	1	1	16384	H	30.501	24
25	S	8	6	N	9/10	2	1	16384	H	26.771	25
26	S	2	2	F	49/50	0	1	4096	H	0.780	26
27	S	2	2	F	49/50	1	1	4096	H	0.785	27
28	S	2	2	F	49/50	2	1	4096	H	0.747	28
29	S	2	2	F	9/10	0	1	16384	H	0.807	29
30	S	2	2	F	9/10	1	1	16384	H	0.809	30
31	S	2	2	F	9/10	2	1	16384	H	0.853	31
32											32

31 P: primary cache; S: signed reference extension. H: target attained; E: declared budget
 32 exhausted. Repetitions 0–2 have identical training and inference seeds, not independent

training draws. All 132 processes and their immutable cumulative driver checkpoints are included. The full JSON retains every per-stage construction, cache, gate and inference clock.

3. ALL FINAL ECONOMIC INTERVALS

TABLE II: Every final actual-cost, gain and target interval

Cohort	d	T	Method	q	Cost	Gain	Target contrast
P	2	2	W	49/50	[0.680, 0.742]	[0.049, 0.094]	[-0.078, -0.033]
P	2	2	W	9/10	[0.674, 0.699]	[0.083, 0.096]	[-0.018, -0.006]
P	2	2	E	49/50	[0.680, 0.742]	[0.049, 0.094]	[-0.078, -0.033]
P	2	2	E	9/10	[0.674, 0.699]	[0.083, 0.096]	[-0.018, -0.006]
P	2	2	Q	49/50	[0.680, 0.742]	[0.049, 0.094]	[-0.078, -0.033]
P	2	2	Q	9/10	[0.674, 0.699]	[0.083, 0.096]	[-0.018, -0.006]
P	2	2	N	49/50	[0.680, 0.742]	[0.049, 0.094]	[-0.078, -0.033]
P	2	2	N	9/10	[0.674, 0.699]	[0.083, 0.096]	[-0.018, -0.006]
P	2	2	F	49/50	[0.680, 0.742]	[0.049, 0.094]	[-0.078, -0.033]
P	2	2	F	9/10	[0.674, 0.699]	[0.083, 0.096]	[-0.018, -0.006]
P	4	4	E	49/50	[1.138, 1.202]	[0.036, 0.102]	[-0.077, -0.011]
P	4	4	E	9/10	[1.138, 1.201]	[0.035, 0.101]	[0.023, 0.088]
P	4	4	Q	49/50	[1.138, 1.202]	[0.036, 0.102]	[-0.077, -0.011]
P	4	4	Q	9/10	[1.138, 1.201]	[0.035, 0.101]	[0.023, 0.088]
P	4	4	N	49/50	[1.138, 1.202]	[0.036, 0.102]	[-0.077, -0.011]
P	4	4	N	9/10	[1.138, 1.201]	[0.035, 0.101]	[0.023, 0.088]
P	8	6	E	49/50	[1.690, 1.711]	[-0.010, 0.010]	[0.024, 0.044]
P	8	6	E	9/10	[1.607, 1.808]	[-0.157, 0.157]	[0.019, 0.323]
P	8	6	Q	49/50	[1.690, 1.711]	[-0.010, 0.010]	[0.024, 0.044]
P	8	6	Q	9/10	[1.607, 1.808]	[-0.157, 0.157]	[0.019, 0.323]
P	8	6	N	49/50	[1.690, 1.711]	[-0.010, 0.010]	[0.024, 0.044]
P	8	6	N	9/10	[1.607, 1.808]	[-0.157, 0.157]	[0.019, 0.323]
S	2	2	W	49/50	[0.533, 0.684]	[0.062, 0.241]	[-0.224, -0.048]
S	2	2	W	9/10	[0.595, 0.651]	[0.129, 0.178]	[-0.098, -0.054]
S	2	2	E	49/50	[0.533, 0.684]	[0.062, 0.241]	[-0.225, -0.048]
S	2	2	E	9/10	[0.595, 0.651]	[0.129, 0.178]	[-0.098, -0.054]

Cohort	d	T	Method	q	Cost	Gain	Target contrast
S	2	2	Q	49/50	[0.533, 0.684]	[0.062, 0.241]	[-0.225, -0.048]
S	2	2	Q	9/10	[0.595, 0.651]	[0.129, 0.178]	[-0.098, -0.054]
S	2	2	N	49/50	[0.533, 0.684]	[0.062, 0.241]	[-0.225, -0.048]
S	2	2	N	9/10	[0.595, 0.651]	[0.129, 0.178]	[-0.098, -0.054]
S	2	2	F	49/50	[0.533, 0.684]	[0.062, 0.241]	[-0.225, -0.048]
S	2	2	F	9/10	[0.595, 0.651]	[0.129, 0.178]	[-0.098, -0.054]
S	4	4	E	49/50	[0.897, 1.066]	[0.119, 0.380]	[-0.354, -0.097]
S	4	4	E	9/10	[0.897, 1.066]	[0.117, 0.378]	[-0.247, -0.003]
S	4	4	Q	49/50	[0.897, 1.066]	[0.119, 0.380]	[-0.354, -0.097]
S	4	4	Q	9/10	[0.897, 1.066]	[0.117, 0.378]	[-0.247, -0.003]
S	4	4	N	49/50	[0.897, 1.066]	[0.119, 0.380]	[-0.354, -0.097]
S	4	4	N	9/10	[0.897, 1.066]	[0.117, 0.378]	[-0.247, -0.003]
S	8	6	E	49/50	[1.342, 1.536]	[0.091, 0.416]	[-0.380, -0.059]
S	8	6	E	9/10	[1.421, 1.479]	[0.214, 0.298]	[-0.125, -0.046]
S	8	6	Q	49/50	[1.342, 1.536]	[0.091, 0.416]	[-0.380, -0.059]
S	8	6	Q	9/10	[1.421, 1.479]	[0.214, 0.298]	[-0.125, -0.046]
S	8	6	N	49/50	[1.342, 1.536]	[0.091, 0.416]	[-0.380, -0.059]
S	8	6	N	9/10	[1.421, 1.479]	[0.214, 0.298]	[-0.125, -0.046]

Intervals are rounded outward and correspond to repetition zero; exact identities across repetitions are separately checked. A negative target-contrast upper endpoint licenses the stated prospective return. Gain intervals use paired path endpoints rather than subtraction of marginal cost intervals. Every intermediate look remains in the service archive.

4. ALL DATEWISE ERROR ACCOUNTS

TABLE III: All datewise directed gaps and enclosure components

Cohort	d	T	Method	q	Stage	Date	Changes	$U - C$	$C - L$	$U - L$
P	2	2	W	49/50	0	0	0	2.195	0.118	2.230

	Cohort	d	T	Method	q	Stage	Date	Changes	$U - C$	$C - L$	$U - L$	
1												1
2		P 2	2	W	49/50	0	1	27	0.607	0.054	0.660	2
3		P 2	2	W	9/10	0	0	0	2.195	0.118	2.230	3
4		P 2	2	W	9/10	0	1	27	0.607	0.054	0.660	4
5		P 2	2	W	9/10	1	0	0	1.481	0.033	1.486	5
6		P 2	2	W	9/10	1	1	93	0.344	0.033	0.377	6
7		P 2	2	E	49/50	0	0	0	2.195	0.118	2.230	7
8		P 2	2	E	49/50	0	1	27	0.607	0.054	0.660	8
9		P 2	2	E	9/10	0	0	0	2.195	0.118	2.230	9
10		P 2	2	E	9/10	0	1	27	0.607	0.054	0.660	10
11		P 2	2	E	9/10	1	0	0	1.481	0.033	1.486	11
12		P 2	2	E	9/10	1	1	93	0.344	0.033	0.377	12
13		P 4	4	E	49/50	0	0	0	4.727	0.062	4.729	13
14		P 4	4	E	49/50	0	1	0	3.705	0.058	3.706	14
15		P 4	4	E	49/50	0	2	0	2.614	0.054	2.615	15
16		P 4	4	E	49/50	0	3	15	0.638	0.054	0.691	16
17		P 4	4	E	49/50	1	0	0	4.530	0.041	4.530	17
18		P 4	4	E	49/50	1	1	0	3.535	0.038	3.536	18
19		P 4	4	E	49/50	1	2	0	2.474	0.035	2.474	19
20		P 4	4	E	49/50	1	3	90	0.636	0.043	0.679	20
21		P 4	4	E	9/10	0	0	0	4.727	0.062	4.729	21
22		P 4	4	E	9/10	0	1	0	3.705	0.058	3.706	22
23		P 4	4	E	9/10	0	2	0	2.614	0.054	2.615	23
24		P 4	4	E	9/10	0	3	15	0.638	0.054	0.691	24
25		P 4	4	E	9/10	1	0	0	4.530	0.041	4.530	25
26		P 4	4	E	9/10	1	1	0	3.535	0.038	3.536	26
27		P 4	4	E	9/10	1	2	0	2.474	0.035	2.474	27
28		P 4	4	E	9/10	1	3	90	0.636	0.043	0.679	28
29		P 8	6	E	49/50	0	0	0	7.047	0.021	7.047	29
30		P 8	6	E	49/50	0	1	0	6.079	0.023	6.079	30
31		P 8	6	E	49/50	0	2	0	5.047	0.024	5.047	31
32		P 8	6	E	49/50	0	3	0	3.946	0.027	3.946	32
		P 8	6	E	49/50	0	4	0	2.771	0.029	2.772	
		P 8	6	E	49/50	0	5	0	0.640	0.074	0.713	

	Cohort	d	T	Method	q	Stage	Date	Changes	$U - C$	$C - L$	$U - L$	
1												1
2		P 8 6		E 49/50	1	0		0	6.836	0.058	6.836	2
3		P 8 6		E 49/50	1	1		0	5.901	0.056	5.902	3
4		P 8 6		E 49/50	1	2		0	4.904	0.055	4.904	4
5		P 8 6		E 49/50	1	3		0	3.840	0.053	3.841	5
6		P 8 6		E 49/50	1	4		0	2.705	0.052	2.706	6
7		P 8 6		E 49/50	1	5		0	0.639	0.064	0.702	7
8		P 8 6		E 9/10	0	0		0	7.047	0.021	7.047	8
9		P 8 6		E 9/10	0	1		0	6.079	0.023	6.079	9
10		P 8 6		E 9/10	0	2		0	5.047	0.024	5.047	10
11		P 8 6		E 9/10	0	3		0	3.946	0.027	3.946	11
12		P 8 6		E 9/10	0	4		0	2.771	0.029	2.772	12
13		P 8 6		E 9/10	0	5		0	0.640	0.074	0.713	13
14		P 8 6		E 9/10	1	0		0	6.836	0.058	6.836	14
15		P 8 6		E 9/10	1	1		0	5.901	0.056	5.902	15
16		P 8 6		E 9/10	1	2		0	4.904	0.055	4.904	16
17		P 8 6		E 9/10	1	3		0	3.840	0.053	3.841	17
18		P 8 6		E 9/10	1	4		0	2.705	0.052	2.706	18
19		P 8 6		E 9/10	1	5		0	0.639	0.064	0.702	19
20		P 2 2		Q 49/50	0	0		0	2.195	0.118	2.230	20
21		P 2 2		Q 49/50	0	1		27	0.607	0.054	0.660	21
22		P 2 2		Q 9/10	0	0		0	2.195	0.118	2.230	22
23		P 2 2		Q 9/10	0	1		27	0.607	0.054	0.660	23
24		P 2 2		Q 9/10	1	0		0	1.361	0.033	1.388	24
25		P 2 2		Q 9/10	1	1		93	0.344	0.033	0.377	25
26		P 4 4		Q 49/50	0	0		0	4.727	0.062	4.729	26
27		P 4 4		Q 49/50	0	1		0	3.705	0.058	3.706	27
28		P 4 4		Q 49/50	0	2		0	2.614	0.054	2.615	28
29		P 4 4		Q 49/50	0	3		15	0.638	0.054	0.691	29
30		P 4 4		Q 49/50	1	0		0	4.530	0.041	4.530	30
31		P 4 4		Q 49/50	1	1		0	3.535	0.038	3.536	31
32		P 4 4		Q 49/50	1	2		0	2.474	0.035	2.474	32
		P 4 4		Q 49/50	1	3		90	0.636	0.043	0.679	
		P 4 4		Q 9/10	0	0		0	4.727	0.062	4.729	

	Cohort	d	T	Method	q	Stage	Date	Changes	$U - C$	$C - L$	$U - L$	
1												1
2		P 4	4	Q	9/10	0	1	0	3.705	0.058	3.706	2
3		P 4	4	Q	9/10	0	2	0	2.614	0.054	2.615	3
4		P 4	4	Q	9/10	0	3	15	0.638	0.054	0.691	4
5		P 4	4	Q	9/10	1	0	0	4.530	0.041	4.530	5
6		P 4	4	Q	9/10	1	1	0	3.535	0.038	3.536	6
7		P 4	4	Q	9/10	1	2	0	2.474	0.035	2.474	7
8		P 4	4	Q	9/10	1	3	90	0.636	0.043	0.679	8
9		P 8	6	Q	49/50	0	0	0	7.047	0.021	7.047	9
10		P 8	6	Q	49/50	0	1	0	6.079	0.023	6.079	10
11		P 8	6	Q	49/50	0	2	0	5.047	0.024	5.047	11
12		P 8	6	Q	49/50	0	3	0	3.946	0.027	3.946	12
13		P 8	6	Q	49/50	0	4	0	2.771	0.029	2.772	13
14		P 8	6	Q	49/50	0	5	0	0.640	0.074	0.713	14
15		P 8	6	Q	49/50	1	0	0	6.836	0.058	6.836	15
16		P 8	6	Q	49/50	1	1	0	5.901	0.056	5.902	16
17		P 8	6	Q	49/50	1	2	0	4.904	0.055	4.904	17
18		P 8	6	Q	49/50	1	3	0	3.840	0.053	3.841	18
19		P 8	6	Q	49/50	1	4	0	2.705	0.052	2.706	19
20		P 8	6	Q	49/50	1	5	0	0.639	0.064	0.702	20
21		P 8	6	Q	9/10	0	0	0	7.047	0.021	7.047	21
22		P 8	6	Q	9/10	0	1	0	6.079	0.023	6.079	22
23		P 8	6	Q	9/10	0	2	0	5.047	0.024	5.047	23
24		P 8	6	Q	9/10	0	3	0	3.946	0.027	3.946	24
25		P 8	6	Q	9/10	0	4	0	2.771	0.029	2.772	25
26		P 8	6	Q	9/10	0	5	0	0.640	0.074	0.713	26
27		P 8	6	Q	9/10	1	0	0	6.836	0.058	6.836	27
28		P 8	6	Q	9/10	1	1	0	5.901	0.056	5.902	28
29		P 8	6	Q	9/10	1	2	0	4.904	0.055	4.904	29
30		P 8	6	Q	9/10	1	3	0	3.840	0.053	3.841	30
31		P 8	6	Q	9/10	1	4	0	2.705	0.052	2.706	31
32		P 8	6	Q	9/10	1	5	0	0.639	0.064	0.702	32
		P 2	2	N	49/50	0	0	0	2.195	0.118	2.230	
		P 2	2	N	49/50	0	1	27	0.607	0.054	0.660	

	Cohort	d	T	Method	q	Stage	Date	Changes	$U - C$	$C - L$	$U - L$	
1												1
2	P	2	2	N	9/10	0	0	0	2.195	0.118	2.230	2
3	P	2	2	N	9/10	0	1	27	0.607	0.054	0.660	3
4	P	2	2	N	9/10	1	0	0	1.481	0.033	1.486	4
5	P	2	2	N	9/10	1	1	93	0.344	0.033	0.377	5
6	P	4	4	N	49/50	0	0	0	4.727	0.062	4.729	6
7	P	4	4	N	49/50	0	1	0	3.705	0.058	3.706	7
8	P	4	4	N	49/50	0	2	0	2.614	0.054	2.615	8
9	P	4	4	N	49/50	0	3	15	0.638	0.054	0.691	9
10	P	4	4	N	49/50	1	0	0	4.530	0.041	4.530	10
11	P	4	4	N	49/50	1	1	0	3.535	0.038	3.536	11
12	P	4	4	N	49/50	1	2	0	2.474	0.035	2.474	12
13	P	4	4	N	49/50	1	3	90	0.636	0.043	0.679	13
14	P	4	4	N	9/10	0	0	0	4.727	0.062	4.729	14
15	P	4	4	N	9/10	0	1	0	3.705	0.058	3.706	15
16	P	4	4	N	9/10	0	2	0	2.614	0.054	2.615	16
17	P	4	4	N	9/10	0	3	15	0.638	0.054	0.691	17
18	P	4	4	N	9/10	1	0	0	4.530	0.041	4.530	18
19	P	4	4	N	9/10	1	1	0	3.535	0.038	3.536	19
20	P	4	4	N	9/10	1	2	0	2.474	0.035	2.474	20
21	P	4	4	N	9/10	1	3	90	0.636	0.043	0.679	21
22	P	8	6	N	49/50	0	0	0	7.047	0.021	7.047	22
23	P	8	6	N	49/50	0	1	0	6.079	0.023	6.079	23
24	P	8	6	N	49/50	0	2	0	5.047	0.024	5.047	24
25	P	8	6	N	49/50	0	3	0	3.946	0.027	3.946	25
26	P	8	6	N	49/50	0	4	0	2.771	0.029	2.772	26
27	P	8	6	N	49/50	0	5	0	0.640	0.074	0.713	27
28	P	8	6	N	49/50	1	0	0	6.836	0.058	6.836	28
29	P	8	6	N	49/50	1	1	0	5.901	0.056	5.902	29
30	P	8	6	N	49/50	1	2	0	4.904	0.055	4.904	30
31	P	8	6	N	49/50	1	3	0	3.840	0.053	3.841	31
32	P	8	6	N	49/50	1	4	0	2.705	0.052	2.706	32
	P	8	6	N	49/50	1	5	0	0.639	0.064	0.702	
	P	8	6	N	9/10	0	0	0	7.047	0.021	7.047	

	Cohort	d	T	Method	q	Stage	Date	Changes	$U - C$	$C - L$	$U - L$		
1												1	
2		P	8	6	N	9/10	0	1	0	6.079	0.023	6.079	2
3		P	8	6	N	9/10	0	2	0	5.047	0.024	5.047	3
4		P	8	6	N	9/10	0	3	0	3.946	0.027	3.946	4
5		P	8	6	N	9/10	0	4	0	2.771	0.029	2.772	5
6		P	8	6	N	9/10	0	5	0	0.640	0.074	0.713	6
7		P	8	6	N	9/10	1	0	0	6.836	0.058	6.836	7
8		P	8	6	N	9/10	1	1	0	5.901	0.056	5.902	8
9		P	8	6	N	9/10	1	2	0	4.904	0.055	4.904	9
10		P	8	6	N	9/10	1	3	0	3.840	0.053	3.841	10
11		P	8	6	N	9/10	1	4	0	2.705	0.052	2.706	11
12		P	8	6	N	9/10	1	5	0	0.639	0.064	0.702	12
13		P	2	2	F	49/50	0	0	0	2.195	0.118	2.230	13
14		P	2	2	F	49/50	0	1	27	0.607	0.054	0.660	14
15		P	2	2	F	9/10	0	0	0	2.195	0.118	2.230	15
16		P	2	2	F	9/10	0	1	27	0.607	0.054	0.660	16
17		P	2	2	F	9/10	1	0	0	1.481	0.033	1.486	17
18		P	2	2	F	9/10	1	1	93	0.344	0.033	0.377	17
18		S	2	2	W	49/50	0	0	30	0.319	0.086	0.405	18
19		S	2	2	W	49/50	0	1	27	0.607	0.054	0.660	19
20		S	2	2	W	9/10	0	0	30	0.319	0.086	0.405	20
21		S	2	2	W	9/10	0	1	27	0.607	0.054	0.660	21
22		S	2	2	E	49/50	0	0	30	0.319	0.086	0.405	22
23		S	2	2	E	49/50	0	1	27	0.607	0.054	0.660	23
24		S	2	2	E	9/10	0	0	30	0.319	0.086	0.405	24
25		S	2	2	E	9/10	0	1	27	0.607	0.054	0.660	24
25		S	4	4	E	49/50	0	0	31	0.503	0.246	0.749	25
26		S	4	4	E	49/50	0	1	31	0.464	0.227	0.691	26
27		S	4	4	E	49/50	0	2	28	0.400	0.197	0.596	27
28		S	4	4	E	49/50	0	3	15	0.638	0.054	0.691	28
29		S	4	4	E	9/10	0	0	31	0.503	0.246	0.749	29
30		S	4	4	E	9/10	0	1	31	0.464	0.227	0.691	30
31		S	4	4	E	9/10	0	2	28	0.400	0.197	0.596	31
32		S	4	4	E	9/10	0	3	15	0.638	0.054	0.691	32

	Cohort	d	T	Method	q	Stage	Date	Changes	$U - C$	$C - L$	$U - L$		
1												1	
2												2	
3		S	8	6	E	49/50	0	0	26	0.604	0.418	1.021	3
4		S	8	6	E	49/50	0	1	24	0.588	0.407	0.994	4
5		S	8	6	E	49/50	0	2	24	0.562	0.388	0.950	5
6		S	8	6	E	49/50	0	3	20	0.521	0.358	0.879	6
7		S	8	6	E	49/50	0	4	14	0.453	0.310	0.762	7
8		S	8	6	E	49/50	0	5	0	0.640	0.074	0.713	8
9		S	8	6	E	9/10	0	0	26	0.604	0.418	1.021	9
10		S	8	6	E	9/10	0	1	24	0.588	0.407	0.994	10
11		S	8	6	E	9/10	0	2	24	0.562	0.388	0.950	11
12		S	8	6	E	9/10	0	3	20	0.521	0.358	0.879	12
13		S	8	6	E	9/10	0	4	14	0.453	0.310	0.762	13
14		S	8	6	E	9/10	0	5	0	0.640	0.074	0.713	14
15		S	2	2	Q	49/50	0	0	30	0.319	0.086	0.405	15
16		S	2	2	Q	49/50	0	1	27	0.607	0.054	0.660	16
17		S	2	2	Q	9/10	0	0	30	0.319	0.086	0.405	17
18		S	2	2	Q	9/10	0	1	27	0.607	0.054	0.660	18
19		S	4	4	Q	49/50	0	0	31	0.503	0.246	0.749	19
20		S	4	4	Q	49/50	0	1	31	0.464	0.227	0.691	20
21		S	4	4	Q	49/50	0	2	28	0.400	0.197	0.596	21
22		S	4	4	Q	49/50	0	3	15	0.638	0.054	0.691	22
23		S	4	4	Q	9/10	0	0	31	0.503	0.246	0.749	23
24		S	4	4	Q	9/10	0	1	31	0.464	0.227	0.691	24
25		S	4	4	Q	9/10	0	2	28	0.400	0.197	0.596	25
26		S	4	4	Q	9/10	0	3	15	0.638	0.054	0.691	26
27		S	8	6	Q	49/50	0	0	26	0.604	0.418	1.021	27
28		S	8	6	Q	49/50	0	1	24	0.588	0.407	0.994	28
29		S	8	6	Q	49/50	0	2	24	0.562	0.388	0.950	29
30		S	8	6	Q	49/50	0	3	20	0.521	0.358	0.879	30
31		S	8	6	Q	49/50	0	4	14	0.453	0.310	0.762	31
32		S	8	6	Q	9/10	0	0	26	0.604	0.418	1.021	32
		S	8	6	Q	9/10	0	1	24	0.588	0.407	0.994	
		S	8	6	Q	9/10	0	2	24	0.562	0.388	0.950	

	Cohort	d	T	Method	q	Stage	Date	Changes	$U - C$	$C - L$	$U - L$	
1												1
2												2
3		S 8 6		Q	9/10	0	3	20	0.521	0.358	0.879	3
4		S 8 6		Q	9/10	0	4	14	0.453	0.310	0.762	4
5		S 8 6		Q	9/10	0	5	0	0.640	0.074	0.713	5
6		S 2 2		N	49/50	0	0	30	0.319	0.086	0.405	6
7		S 2 2		N	49/50	0	1	27	0.607	0.054	0.660	7
8		S 2 2		N	9/10	0	0	30	0.319	0.086	0.405	8
9		S 2 2		N	9/10	0	1	27	0.607	0.054	0.660	9
10		S 4 4		N	49/50	0	0	31	0.503	0.246	0.749	10
11		S 4 4		N	49/50	0	1	31	0.464	0.227	0.691	11
12		S 4 4		N	49/50	0	2	28	0.400	0.197	0.596	12
13		S 4 4		N	49/50	0	3	15	0.638	0.054	0.691	13
14		S 4 4		N	9/10	0	0	31	0.503	0.246	0.749	14
15		S 4 4		N	9/10	0	1	31	0.464	0.227	0.691	15
16		S 4 4		N	9/10	0	2	28	0.400	0.197	0.596	16
17		S 4 4		N	9/10	0	3	15	0.638	0.054	0.691	17
18		S 8 6		N	49/50	0	0	26	0.604	0.418	1.021	18
19		S 8 6		N	49/50	0	1	24	0.588	0.407	0.994	19
20		S 8 6		N	49/50	0	2	24	0.562	0.388	0.950	20
21		S 8 6		N	49/50	0	3	20	0.521	0.358	0.879	21
22		S 8 6		N	49/50	0	4	14	0.453	0.310	0.762	22
23		S 8 6		N	49/50	0	5	0	0.640	0.074	0.713	23
24		S 8 6		N	9/10	0	0	26	0.604	0.418	1.021	24
25		S 8 6		N	9/10	0	1	24	0.588	0.407	0.994	25
26		S 8 6		N	9/10	0	2	24	0.562	0.388	0.950	26
27		S 8 6		N	9/10	0	3	20	0.521	0.358	0.879	27
28		S 8 6		N	9/10	0	4	14	0.453	0.310	0.762	28
29		S 8 6		N	9/10	0	5	0	0.640	0.074	0.713	29
30		S 2 2		F	49/50	0	0	30	0.319	0.086	0.405	30
31		S 2 2		F	49/50	0	1	27	0.607	0.054	0.660	31
32		S 2 2		F	9/10	0	0	30	0.319	0.086	0.405	32
		S 2 2		F	9/10	0	1	27	0.607	0.054	0.660	

All stages of repetition zero are retained. A maximum of sums need not equal the sum of maxima. Quantities are upper bounds rounded upward, not estimates of disjoint causal errors. Complete cell arrays contain the exact endpoints used by the gate.

5. READING AND PRESERVATION MAP

The active article is organized around construction and training, acquired-state improvement, prospective stopping and actual economic cost. The preceding main article is retained as an unchanged historical source and a rebuilt development document. The complete development article and complete proof supplement retain the full prior body and add the current theoretical and numerical sections. Historical labels and immutable section-source hashes are checked rather than relying on a statement that no results were removed.

The original resource-allocation, original-owner compiler, observation-price, recursive-preference, diffusion, temporal-self and game results remain available in those companions. The expected-cost sampling and linear action-kernel results in the active article do not claim that a nonlinear recursive utility or game has been experimentally validated by an investment-policy interval. A reading map is a distinction among assumptions and evidence, not abandonment of the original NBO research object.

6. EXECUTED ACTION-SEARCH SERVICES AND COMPLETE NUMERICAL ACCOUNTS

This section records the new R57 execution, separately from the preceding R55 cohorts and their R56 integration. Its scientific protocol, executable source and declared target family were frozen before production. The publication replay does not retrain the critics, choose new seeds, draw new economic observations or replace the original service clocks.

6.1. *Exact search and proposal identity*

For each saved critic, the replay reconstructs the integrated action objective from the exact stored dyadic coefficients. The action-dependent objective drops only a common additive constant. It recomputes rational hinge breakpoints, cubic stationary-root isolation and exact rational comparisons on the action lattice. It

checks the chosen index, objective value, nonterminal menu optimization loss and root/piece counts. Different valid rational isolating brackets are not by themselves different economic actions; the selected lattice index and exact objective are the controlling solver identities.

The ReLU-exact mode keeps every ReLU-menu candidate. The quadratic mode has its own exact fitted-value search; the other modes retain their declared menu construction. Every terminal proposal uses the same analytic terminal objective under the original law. The whole-cell signed reference verifier is independent of the critic, so a shared action receives identical endpoints in the matched search comparison. The replay reconstructs all raw candidate arrays, selected actions, continuous-action lower covers and datewise gaps. Removing the added algebraic action uses the stored pre-addition minimum and policy; no different training run is substituted as the counterfactual menu.

6.2. Every target-service return

TABLE V: Every prospective target-service return

Service	Attempts	Leaves	Paths	Work (s)	Reached
2/2-20-C-1	2	64	131072	3.776	yes
2/2-20-C-2	2	64	131072	3.793	yes
2/2-20-C-3	2	64	131072	3.763	yes
2/2-10-C-1	1	32	16384	1.248	yes
2/2-10-C-2	1	32	16384	1.242	yes
2/2-10-C-3	1	32	16384	1.220	yes
4/4-20-C-1	2	64	131072	12.983	yes
4/4-20-C-2	2	64	131072	12.881	yes
4/4-20-C-3	2	64	131072	12.439	yes
4/4-10-C-1	1	32	4096	3.363	yes
4/4-10-C-2	1	32	4096	3.382	yes
4/4-10-C-3	1	32	4096	3.477	yes
8/6-20-C-1	2	64	81920	35.571	no
8/6-20-C-2	2	64	81920	35.049	no

Every prospective target-service return (continued)

Service	Attempts	Leaves	Paths	Work (s)	Reached
8/6-20-C-3	2	64	81920	35.153	no
8/6-10-C-1	1	32	16384	11.290	yes
8/6-10-C-2	1	32	16384	11.470	yes
8/6-10-C-3	1	32	16384	11.337	yes
2/2-20-E-1	2	64	131072	5.202	yes
2/2-20-E-2	2	64	131072	5.168	yes
2/2-20-E-3	2	64	131072	5.185	yes
2/2-10-E-1	1	32	16384	2.075	yes
2/2-10-E-2	1	32	16384	1.954	yes
2/2-10-E-3	1	32	16384	1.928	yes
4/4-20-E-1	2	64	131072	15.872	yes
4/4-20-E-2	2	64	131072	16.513	yes
4/4-20-E-3	2	64	131072	15.984	yes
4/4-10-E-1	1	32	4096	4.665	yes
4/4-10-E-2	1	32	4096	4.583	yes
4/4-10-E-3	1	32	4096	4.626	yes
8/6-20-E-1	2	64	81920	40.704	no
8/6-20-E-2	2	64	81920	40.766	no
8/6-20-E-3	2	64	81920	41.016	no
8/6-10-E-1	1	32	16384	13.589	yes
8/6-10-E-2	1	32	16384	13.204	yes
8/6-10-E-3	1	32	16384	13.115	yes
2/2-20-Q-1	2	64	131072	4.107	yes
2/2-20-Q-2	2	64	131072	4.043	yes
2/2-20-Q-3	2	64	131072	4.083	yes
2/2-10-Q-1	1	32	16384	1.350	yes
2/2-10-Q-2	1	32	16384	1.340	yes
2/2-10-Q-3	1	32	16384	1.324	yes
4/4-20-Q-1	2	64	131072	13.991	yes
4/4-20-Q-2	2	64	131072	13.475	yes
4/4-20-Q-3	2	64	131072	13.780	yes
4/4-10-Q-1	1	32	4096	3.753	yes

Every prospective target-service return (continued)						
Service	Attempts	Leaves	Paths	Work (s)	Reached	
4/4-10-Q-2	1	32	4096	3.698	yes	
4/4-10-Q-3	1	32	4096	3.713	yes	
8/6-20-Q-1	2	64	81920	38.954	no	
8/6-20-Q-2	2	64	81920	38.257	no	
8/6-20-Q-3	2	64	81920	38.682	no	
8/6-10-Q-1	1	32	16384	12.302	yes	
8/6-10-Q-2	1	32	16384	12.451	yes	
8/6-10-Q-3	1	32	16384	12.564	yes	
2/2-20-X-1	2	64	131072	5.238	yes	
2/2-20-X-2	2	64	131072	5.179	yes	
2/2-20-X-3	2	64	131072	5.216	yes	
2/2-10-X-1	1	32	16384	1.700	yes	
2/2-10-X-2	1	32	16384	1.666	yes	
2/2-10-X-3	1	32	16384	1.697	yes	
4/4-20-X-1	2	64	131072	17.786	yes	
4/4-20-X-2	2	64	131072	17.951	yes	
4/4-20-X-3	2	64	131072	18.369	yes	
4/4-10-X-1	1	32	4096	5.272	yes	
4/4-10-X-2	1	32	4096	4.865	yes	
4/4-10-X-3	1	32	4096	5.122	yes	
8/6-20-X-1	2	64	81920	44.183	no	
8/6-20-X-2	2	64	81920	45.418	no	
8/6-20-X-3	2	64	81920	44.189	no	
8/6-10-X-1	1	32	16384	14.165	yes	
8/6-10-X-2	1	32	16384	13.960	yes	
8/6-10-X-3	1	32	16384	14.161	yes	
2/2-20-N-1	2	64	131072	4.511	yes	
2/2-20-N-2	2	64	131072	4.536	yes	
2/2-20-N-3	2	64	131072	4.497	yes	
2/2-10-N-1	1	32	16384	1.471	yes	
2/2-10-N-2	1	32	16384	1.435	yes	
2/2-10-N-3	1	32	16384	1.458	yes	

Every prospective target-service return (continued)

Service	Attempts	Leaves	Paths	Work (s)	Reached
4/4-20-N-1	2	64	131072	15.004	yes
4/4-20-N-2	2	64	131072	15.240	yes
4/4-20-N-3	2	64	131072	15.077	yes
4/4-10-N-1	1	32	4096	3.929	yes
4/4-10-N-2	1	32	4096	3.963	yes
4/4-10-N-3	1	32	4096	3.949	yes
8/6-20-N-1	2	64	81920	39.190	no
8/6-20-N-2	2	64	81920	39.458	no
8/6-20-N-3	2	64	81920	39.239	no
8/6-10-N-1	1	32	16384	12.484	yes
8/6-10-N-2	1	32	16384	12.188	yes
8/6-10-N-3	1	32	16384	12.581	yes

Service identifiers encode dimension/horizon, percent-reduction target, mode, and seed index 1–3.
 Paths sum the final cumulative look in each attempted construction; nested looks are not double-counted.
 All 90 services are retained, including every budget exhaustion.

TABLE VI: Every returned policy and its original Bellman account

Service	Expected cost	Gain from installed	All-state bound
2/2-20-C-1	[0.59687, 0.61854]	[0.15900, 0.17347]	3.85741
2/2-20-C-2	[0.59646, 0.61815]	[0.15944, 0.17391]	3.85741
2/2-20-C-3	[0.59766, 0.61949]	[0.15938, 0.17388]	3.85741
2/2-10-C-1	[0.59456, 0.64992]	[0.12935, 0.17762]	3.94572
2/2-10-C-2	[0.59742, 0.65311]	[0.12966, 0.17792]	3.94572
2/2-10-C-3	[0.59889, 0.65459]	[0.13026, 0.17850]	3.94572
4/4-20-C-1	[0.95342, 0.97249]	[0.26227, 0.28146]	6.72215
4/4-20-C-2	[0.95352, 0.97261]	[0.26187, 0.28108]	6.72215
4/4-20-C-3	[0.95430, 0.97342]	[0.26221, 0.28144]	6.72215
4/4-10-C-1	[0.90014, 1.06847]	[0.11978, 0.37950]	6.78416
4/4-10-C-2	[0.89574, 1.06400]	[0.11750, 0.37711]	6.78416

Every returned policy and its original Bellman account (continued)				
Service	Expected cost	Gain from installed	All-state bound	
4/4-10-C-3	[0.90078, 1.06845]	[0.12085, 0.38038]	6.78416	
8/6-20-C-1	[1.34699, 1.36544]	[0.33397, 0.35624]	9.18783	
8/6-20-C-2	[1.34634, 1.36475]	[0.33372, 0.35598]	9.18783	
8/6-20-C-3	[1.34640, 1.36482]	[0.33384, 0.35609]	9.18783	
8/6-10-C-1	[1.41623, 1.47339]	[0.21298, 0.29622]	9.24880	
8/6-10-C-2	[1.42016, 1.47724]	[0.21348, 0.29669]	9.24880	
8/6-10-C-3	[1.41810, 1.47521]	[0.21329, 0.29649]	9.24880	
2/2-20-E-1	[0.59681, 0.61848]	[0.15905, 0.17353]	3.85741	
2/2-20-E-2	[0.59636, 0.61805]	[0.15954, 0.17401]	3.85741	
2/2-20-E-3	[0.59780, 0.61962]	[0.15924, 0.17374]	3.85741	
2/2-10-E-1	[0.59456, 0.64992]	[0.12935, 0.17762]	3.94572	
2/2-10-E-2	[0.59742, 0.65311]	[0.12966, 0.17792]	3.94572	
2/2-10-E-3	[0.59889, 0.65459]	[0.13026, 0.17850]	3.94572	
4/4-20-E-1	[0.95347, 0.97254]	[0.26221, 0.28141]	6.72215	
4/4-20-E-2	[0.95352, 0.97261]	[0.26187, 0.28108]	6.72215	
4/4-20-E-3	[0.95430, 0.97342]	[0.26221, 0.28144]	6.72215	
4/4-10-E-1	[0.90014, 1.06847]	[0.11978, 0.37950]	6.78416	
4/4-10-E-2	[0.89574, 1.06400]	[0.11750, 0.37711]	6.78416	
4/4-10-E-3	[0.90062, 1.06826]	[0.12102, 0.38056]	6.78416	
8/6-20-E-1	[1.34699, 1.36544]	[0.33397, 0.35624]	9.18783	
8/6-20-E-2	[1.34626, 1.36467]	[0.33380, 0.35606]	9.18783	
8/6-20-E-3	[1.34627, 1.36469]	[0.33397, 0.35621]	9.18783	
8/6-10-E-1	[1.41623, 1.47339]	[0.21298, 0.29622]	9.24880	
8/6-10-E-2	[1.42016, 1.47724]	[0.21348, 0.29669]	9.24880	
8/6-10-E-3	[1.41810, 1.47521]	[0.21329, 0.29649]	9.24880	
2/2-20-Q-1	[0.59687, 0.61854]	[0.15900, 0.17347]	3.85741	
2/2-20-Q-2	[0.59646, 0.61815]	[0.15944, 0.17391]	3.85741	
2/2-20-Q-3	[0.59766, 0.61949]	[0.15938, 0.17388]	3.85741	
2/2-10-Q-1	[0.59456, 0.64992]	[0.12935, 0.17762]	3.94572	
2/2-10-Q-2	[0.59742, 0.65311]	[0.12966, 0.17792]	3.94572	
2/2-10-Q-3	[0.59889, 0.65459]	[0.13026, 0.17850]	3.94572	
4/4-20-Q-1	[0.95342, 0.97249]	[0.26227, 0.28146]	6.72215	

Every returned policy and its original Bellman account (continued)				
Service	Expected cost	Gain from installed	All-state bound	
4/4-20-Q-2	[0.95352, 0.97261]	[0.26187, 0.28108]	6.72215	
4/4-20-Q-3	[0.95430, 0.97342]	[0.26221, 0.28144]	6.72215	
4/4-10-Q-1	[0.90014, 1.06847]	[0.11978, 0.37950]	6.78416	
4/4-10-Q-2	[0.89574, 1.06400]	[0.11750, 0.37711]	6.78416	
4/4-10-Q-3	[0.90078, 1.06845]	[0.12085, 0.38038]	6.78416	
8/6-20-Q-1	[1.34699, 1.36544]	[0.33397, 0.35624]	9.18783	
8/6-20-Q-2	[1.34634, 1.36475]	[0.33372, 0.35598]	9.18783	
8/6-20-Q-3	[1.34640, 1.36482]	[0.33384, 0.35609]	9.18783	
8/6-10-Q-1	[1.41623, 1.47339]	[0.21298, 0.29622]	9.24880	
8/6-10-Q-2	[1.42016, 1.47724]	[0.21348, 0.29669]	9.24880	
8/6-10-Q-3	[1.41810, 1.47521]	[0.21329, 0.29649]	9.24880	
2/2-20-X-1	[0.59688, 0.61856]	[0.15898, 0.17346]	3.85741	
2/2-20-X-2	[0.59648, 0.61817]	[0.15942, 0.17389]	3.85741	
2/2-20-X-3	[0.59765, 0.61948]	[0.15939, 0.17389]	3.85741	
2/2-10-X-1	[0.59456, 0.64992]	[0.12935, 0.17762]	3.94572	
2/2-10-X-2	[0.59742, 0.65311]	[0.12966, 0.17792]	3.94572	
2/2-10-X-3	[0.59892, 0.65461]	[0.13023, 0.17848]	3.94572	
4/4-20-X-1	[0.95335, 0.97242]	[0.26234, 0.28152]	6.72215	
4/4-20-X-2	[0.95346, 0.97255]	[0.26193, 0.28114]	6.72215	
4/4-20-X-3	[0.95430, 0.97342]	[0.26221, 0.28144]	6.72215	
4/4-10-X-1	[0.90014, 1.06847]	[0.11978, 0.37950]	6.78416	
4/4-10-X-2	[0.89574, 1.06400]	[0.11750, 0.37711]	6.78416	
4/4-10-X-3	[0.90078, 1.06845]	[0.12085, 0.38038]	6.78416	
8/6-20-X-1	[1.34699, 1.36544]	[0.33397, 0.35624]	9.18783	
8/6-20-X-2	[1.34634, 1.36475]	[0.33372, 0.35598]	9.18783	
8/6-20-X-3	[1.34640, 1.36482]	[0.33384, 0.35609]	9.18783	
8/6-10-X-1	[1.41623, 1.47339]	[0.21298, 0.29622]	9.24880	
8/6-10-X-2	[1.42016, 1.47724]	[0.21348, 0.29669]	9.24880	
8/6-10-X-3	[1.41810, 1.47521]	[0.21329, 0.29649]	9.24880	
2/2-20-N-1	[0.59687, 0.61854]	[0.15900, 0.17347]	3.85741	
2/2-20-N-2	[0.59646, 0.61815]	[0.15944, 0.17391]	3.85741	
2/2-20-N-3	[0.59766, 0.61949]	[0.15938, 0.17388]	3.85741	

Every returned policy and its original Bellman account (continued)				
Service	Expected cost	Gain from installed	All-state bound	
2/2-10-N-1	[0.59456, 0.64992]	[0.12935, 0.17762]	3.94572	
2/2-10-N-2	[0.59742, 0.65311]	[0.12966, 0.17792]	3.94572	
2/2-10-N-3	[0.59889, 0.65459]	[0.13026, 0.17850]	3.94572	
4/4-20-N-1	[0.95335, 0.97242]	[0.26234, 0.28152]	6.72215	
4/4-20-N-2	[0.95345, 0.97254]	[0.26194, 0.28114]	6.72215	
4/4-20-N-3	[0.95430, 0.97342]	[0.26221, 0.28144]	6.72215	
4/4-10-N-1	[0.90014, 1.06847]	[0.11978, 0.37950]	6.78416	
4/4-10-N-2	[0.89574, 1.06400]	[0.11750, 0.37711]	6.78416	
4/4-10-N-3	[0.90078, 1.06845]	[0.12085, 0.38038]	6.78416	
8/6-20-N-1	[1.34699, 1.36544]	[0.33397, 0.35624]	9.18783	
8/6-20-N-2	[1.34634, 1.36475]	[0.33372, 0.35598]	9.18783	
8/6-20-N-3	[1.34640, 1.36482]	[0.33384, 0.35609]	9.18783	
8/6-10-N-1	[1.41623, 1.47339]	[0.21298, 0.29622]	9.24880	
8/6-10-N-2	[1.42016, 1.47724]	[0.21348, 0.29669]	9.24880	
8/6-10-N-3	[1.41810, 1.47521]	[0.21329, 0.29649]	9.24880	

Intervals are outward rounded. The last column is the one-sweep policy-loss bound against the original Bellman optimum, including the old-continuation term, not merely the local greedy gap. The expected-cost target and this conservative uniform bound are different quantities. Each service is a separate from-primitives process with its own inference and durable output. Within a service, resource refinement restarts from the same installed reference, and no intermediate candidate is deployed. A target-attained return requires a nonpositive confidence upper bound on the signed target estimand. A budget-exhausted return carries a statewise-safe candidate but no attained-target claim. The registry preserves the complete declared set rather than selecting successful seeds.

The all-state column uses the support-capped one-sweep recursion against the original Bellman optimum. It is not the directed greedy gap alone. The old continuation error remains part of the account even when every selected action is verified nonworsening. The exact finite-sweep theorem retains its own premises; two re-

TABLE IV
PROSPECTIVE TEN-PERCENT TARGET SERVICES

d/T	Mode	Reached	Cost upper	Median (s)	Range (s)
2/2	C	3/3	0.65459	1.242	1.220–1.248
2/2	N	3/3	0.65459	1.458	1.435–1.471
2/2	X	3/3	0.65461	1.697	1.666–1.700
2/2	Q	3/3	0.65459	1.340	1.324–1.350
2/2	E	3/3	0.65459	1.954	1.928–2.075
4/4	C	3/3	1.06847	3.382	3.363–3.477
4/4	N	3/3	1.06847	3.949	3.929–3.963
4/4	X	3/3	1.06847	5.122	4.865–5.272
4/4	Q	3/3	1.06847	3.713	3.698–3.753
4/4	E	3/3	1.06847	4.626	4.583–4.665
8/6	C	3/3	1.47724	11.337	11.290–11.470
8/6	N	3/3	1.47724	12.484	12.188–12.581
8/6	X	3/3	1.47724	14.161	13.960–14.165
8/6	Q	3/3	1.47724	12.451	12.302–12.564
8/6	E	3/3	1.47724	13.204	13.115–13.589

C uses common candidates without fitting. N uses the trained ReLU menu; X adds its exact algebraic witness; Q uses quadratic exact search; E uses ExtraTrees menu proposals. All modes receive the same exact terminal action. Three different training seeds are reconstructed separately. Cost upper is the largest returned-policy upper endpoint across those seeds, not a confidence bound for a population mean. Complete process times include every own failed attempt and inference look.

source refinements against a fixed installed reference are not two simultaneous policy-iteration passes against successive incumbents.

6.3. Independent returned-policy comparisons

TABLE VII: Every independent validation contrast

d/T	Target	Seed	X minus	Cost difference	Conclusion
2/2	20	1 C		[-0.00513, 0.00516]	unresolved

	Every independent validation contrast (continued)					
	d/T	Target	Seed	X minus	Cost difference	Conclusion
1	2/2	20	1	E	[-0.00508, 0.00522]	unresolved
2	2/2	20	1	Q	[-0.00513, 0.00516]	unresolved
3	2/2	20	1	N	[-0.00513, 0.00516]	unresolved
4	2/2	20	2	C	[-0.00513, 0.00517]	unresolved
5	2/2	20	2	E	[-0.00504, 0.00528]	unresolved
6	2/2	20	2	Q	[-0.00513, 0.00517]	unresolved
7	2/2	20	2	N	[-0.00513, 0.00516]	unresolved
8	2/2	20	3	C	[-0.00515, 0.00513]	unresolved
9	2/2	20	3	E	[-0.00529, 0.00501]	unresolved
10	2/2	20	3	Q	[-0.00515, 0.00513]	unresolved
11	2/2	20	3	N	[-0.00515, 0.00513]	unresolved
12	2/2	10	1	C	[0.00000, 0.00000]	identity
13	2/2	10	1	E	[0.00000, 0.00000]	identity
14	2/2	10	1	Q	[0.00000, 0.00000]	identity
15	2/2	10	1	N	[0.00000, 0.00000]	identity
16	2/2	10	2	C	[0.00000, 0.00000]	identity
17	2/2	10	2	E	[0.00000, 0.00000]	identity
18	2/2	10	2	Q	[0.00000, 0.00000]	identity
19	2/2	10	2	N	[0.00000, 0.00000]	identity
20	2/2	10	3	C	[-0.00512, 0.00517]	unresolved
21	2/2	10	3	E	[-0.00512, 0.00517]	unresolved
22	2/2	10	3	Q	[-0.00512, 0.00517]	unresolved
23	2/2	10	3	N	[-0.00512, 0.00517]	unresolved
24	4/4	20	1	C	[-0.00772, 0.00758]	unresolved
25	4/4	20	1	E	[-0.00779, 0.00753]	unresolved
26	4/4	20	1	Q	[-0.00772, 0.00758]	unresolved
27	4/4	20	1	N	[0.00000, 0.00000]	identity
28	4/4	20	2	C	[-0.00771, 0.00758]	unresolved
29	4/4	20	2	E	[-0.00771, 0.00758]	unresolved
30	4/4	20	2	Q	[-0.00771, 0.00758]	unresolved
31	4/4	20	2	N	[-0.00763, 0.00764]	unresolved
32	4/4	20	3	C	[-0.00763, 0.00763]	unresolved

Every independent validation contrast (continued)

d/T	Target	Seed	X minus	Cost difference	Conclusion
4/4	20	3 E		[-0.00763, 0.00763]	unresolved
4/4	20	3 Q		[-0.00763, 0.00763]	unresolved
4/4	20	3 N		[-0.00763, 0.00763]	unresolved
4/4	10	1 C		[0.00000, 0.00000]	identity
4/4	10	1 E		[0.00000, 0.00000]	identity
4/4	10	1 Q		[0.00000, 0.00000]	identity
4/4	10	1 N		[0.00000, 0.00000]	identity
4/4	10	2 C		[0.00000, 0.00000]	identity
4/4	10	2 E		[0.00000, 0.00000]	identity
4/4	10	2 Q		[0.00000, 0.00000]	identity
4/4	10	2 N		[0.00000, 0.00000]	identity
4/4	10	3 C		[0.00000, 0.00000]	identity
4/4	10	3 E		[-0.00748, 0.00782]	unresolved
4/4	10	3 Q		[0.00000, 0.00000]	identity
4/4	10	3 N		[0.00000, 0.00000]	identity
8/6	20	1 C		[0.00000, 0.00000]	identity
8/6	20	1 E		[0.00000, 0.00000]	identity
8/6	20	1 Q		[0.00000, 0.00000]	identity
8/6	20	1 N		[0.00000, 0.00000]	identity
8/6	20	2 C		[0.00000, 0.00000]	identity
8/6	20	2 E		[-0.00975, 0.00990]	unresolved
8/6	20	2 Q		[0.00000, 0.00000]	identity
8/6	20	2 N		[0.00000, 0.00000]	identity
8/6	20	3 C		[0.00000, 0.00000]	identity
8/6	20	3 E		[-0.00973, 0.00996]	unresolved
8/6	20	3 Q		[0.00000, 0.00000]	identity
8/6	20	3 N		[0.00000, 0.00000]	identity
8/6	10	1 C		[0.00000, 0.00000]	identity
8/6	10	1 E		[0.00000, 0.00000]	identity
8/6	10	1 Q		[0.00000, 0.00000]	identity
8/6	10	1 N		[0.00000, 0.00000]	identity
8/6	10	2 C		[0.00000, 0.00000]	identity

Every independent validation contrast (continued)

d/T	Target	Seed	X minus	Cost difference	Conclusion
8/6	10	2	E	[0.00000, 0.00000]	identity
8/6	10	2	Q	[0.00000, 0.00000]	identity
8/6	10	2	N	[0.00000, 0.00000]	identity
8/6	10	3	C	[0.00000, 0.00000]	identity
8/6	10	3	E	[0.00000, 0.00000]	identity
8/6	10	3	Q	[0.00000, 0.00000]	identity
8/6	10	3	N	[0.00000, 0.00000]	identity

Each nonidentity interval comes from 65,536 fresh continuous-bin paths for the five returned policies. Negative values favor X. Full raw endpoint arrays, stream hashes, fitted models and prospective stopping records are deposited. The validation stream is generated only after the corresponding target services have returned. The replay reconstructs its original forty-bit initial-state and innovation bins and recomputes every path-cost enclosure. It likewise reconstructs every original stopping stream and every declared look, including unsuccessful early looks. Replaying the same frozen stream is verification, not an increase in sample size.

Original confidence records are verified using exact rational first and second sample moments and the inherited rounded-moment bounds. This separates an arithmetic guarantee from platform-dependent floating-point reduction order. The support for a signed difference includes negative values. Statewise expected safety relative to the installed policy is not used to clip common-shock path gains. Exact policy identities and unresolved statistical differences are retained as different cases.

The new confidence family bounds all declared stopping and paired intervals simultaneously. It is not a claim of independence across methods, targets or reused policy arrays. The three different training seeds generate different fitted objects; they do not establish an optimizer-success distribution or generalization across an unobserved population of economic environments.

6.4. Work, storage and the recorded timing boundary

TABLE VIII: Recorded work components at the twenty-percent target

d/T	Mode	Fit (s)	Search (s)	Verify (s)	Infer (s)	Bytes	RSS (KiB)
2/2	C	0.025	0.124	1.522	1.644	5389008	113044
2/2	N	0.629	0.233	1.525	1.652	5685048	131260
2/2	X	0.626	0.828	1.624	1.659	5729943	130796
2/2	Q	0.095	0.239	1.622	1.660	5706573	113044
2/2	E	1.057	0.325	1.554	1.678	6768970	161412
4/4	C	0.020	0.112	8.649	3.632	5423386	116776
4/4	N	2.417	0.391	8.191	3.584	6345224	133160
4/4	X	2.446	2.705	8.715	3.590	6494162	134948
4/4	Q	0.281	0.471	8.878	3.617	6405642	116776
4/4	E	2.791	0.674	8.257	3.593	8949247	164792
8/6	C	0.018	0.101	30.398	4.264	3531723	117020
8/6	N	3.214	0.530	30.653	4.313	5899194	133432
8/6	X	3.191	4.336	31.978	4.338	6148289	136324
8/6	Q	0.828	0.928	32.146	4.306	5967697	117020
8/6	E	4.192	0.947	30.614	4.281	10024709	168828

Components sum within a run, but their separate medians need not sum to the median complete process time. They exclude some serialization and process overhead that is included in the complete clock. Bytes and RSS are maxima across the three runs. Cross-task runners are not treated as a controlled hardware-scaling regression. The complete own-service clock surrounds the subprocess through termination and durable service records. Internal components separately measure fitting, algebraic search, safety verification and inference through endpoint serialization. Exact rational search pays for its coefficient arithmetic, stationary-root isolation and candidate comparisons. Logical operation counts do not replace the measured process cost, and process seconds are not claimed to be machine independent.

The supplementary paired-comparison timer is narrower: it ends after endpoint computation and serialization, before the final comparison JSON is committed. The driver executes this comparison in its parent process. Thus the intended sep-

arate complete comparison-process timer in the protocol was not obtained. The recorded value is a partial, additional study-overhead measurement, not a complete deployment clock. All own-target-service clocks remain complete. The publication preserves and labels the narrower timing rather than inventing an unobserved overhead or silently amending the raw record.

6.5. Datewise decomposition and preservation

TABLE IX: Datewise verification for the first declared seed

Service	Attempt	Date	Changed	Extra	$U - C$	$C - L$	$U - L$
2/2-20-C-1	1	0	30	0	0.31866	0.08581	0.40446
2/2-20-C-1	1	1	27	0	0.60626	0.05340	0.65965
2/2-20-C-1	2	0	62	0	0.26553	0.05781	0.31615
2/2-20-C-1	2	1	58	0	0.45576	0.04002	0.49577
4/4-20-C-1	1	0	31	0	0.50284	0.24561	0.74845
4/4-20-C-1	1	1	31	0	0.46336	0.22679	0.69015
4/4-20-C-1	1	2	28	0	0.39956	0.19612	0.59568
4/4-20-C-1	1	3	15	0	0.63704	0.05329	0.69033
4/4-20-C-1	2	0	63	0	0.47865	0.20779	0.68644
4/4-20-C-1	2	1	63	0	0.43964	0.19181	0.63144
4/4-20-C-1	2	2	59	0	0.37687	0.16586	0.54273
4/4-20-C-1	2	3	42	0	0.63602	0.04785	0.68386
8/6-20-C-1	1	0	26	0	0.60302	0.41769	1.02070
8/6-20-C-1	1	1	24	0	0.58712	0.40623	0.99334
8/6-20-C-1	1	2	24	0	0.56153	0.38781	0.94934
8/6-20-C-1	1	3	20	0	0.52013	0.35796	0.87809
8/6-20-C-1	1	4	14	0	0.45268	0.30907	0.76175
8/6-20-C-1	1	5	0	0	0.63956	0.07309	0.71264
8/6-20-C-1	2	0	60	0	0.58520	0.37453	0.95973
8/6-20-C-1	2	1	60	0	0.56954	0.36417	0.93371
8/6-20-C-1	2	2	59	0	0.54452	0.34754	0.89205
8/6-20-C-1	2	3	54	0	0.50409	0.32066	0.82475
8/6-20-C-1	2	4	46	0	0.43849	0.27676	0.71525

Datewise verification for the first declared seed (continued)								
Service	Attempt	Date	Changed	Extra	$U - C$	$C - L$	$U - L$	
8/6-20-C-1	2	5	0	0	0.63853	0.06811	0.70663	
2/2-20-E-1	1	0	30	0	0.31866	0.08581	0.40446	
2/2-20-E-1	1	1	27	0	0.60626	0.05340	0.65965	
2/2-20-E-1	2	0	62	0	0.26553	0.05781	0.31615	
2/2-20-E-1	2	1	58	0	0.45576	0.04002	0.49577	
4/4-20-E-1	1	0	31	0	0.50284	0.24561	0.74845	
4/4-20-E-1	1	1	31	0	0.46336	0.22679	0.69015	
4/4-20-E-1	1	2	28	0	0.39956	0.19612	0.59568	
4/4-20-E-1	1	3	15	0	0.63704	0.05329	0.69033	
4/4-20-E-1	2	0	63	0	0.47865	0.20779	0.68644	
4/4-20-E-1	2	1	63	0	0.43964	0.19181	0.63144	
4/4-20-E-1	2	2	59	0	0.37687	0.16586	0.54273	
4/4-20-E-1	2	3	42	0	0.63602	0.04785	0.68386	
8/6-20-E-1	1	0	26	0	0.60302	0.41769	1.02070	
8/6-20-E-1	1	1	24	0	0.58712	0.40623	0.99334	
8/6-20-E-1	1	2	24	0	0.56153	0.38781	0.94934	
8/6-20-E-1	1	3	20	0	0.52013	0.35796	0.87809	
8/6-20-E-1	1	4	14	0	0.45268	0.30907	0.76175	
8/6-20-E-1	1	5	0	0	0.63956	0.07309	0.71264	
8/6-20-E-1	2	0	60	0	0.58520	0.37453	0.95973	
8/6-20-E-1	2	1	60	0	0.56954	0.36417	0.93371	
8/6-20-E-1	2	2	59	0	0.54452	0.34754	0.89205	
8/6-20-E-1	2	3	54	0	0.50409	0.32066	0.82475	
8/6-20-E-1	2	4	46	0	0.43849	0.27676	0.71525	
8/6-20-E-1	2	5	0	0	0.63853	0.06811	0.70663	
2/2-20-Q-1	1	0	30	0	0.31866	0.08581	0.40446	
2/2-20-Q-1	1	1	27	0	0.60626	0.05340	0.65965	
2/2-20-Q-1	2	0	62	0	0.26553	0.05781	0.31615	
2/2-20-Q-1	2	1	58	0	0.45576	0.04002	0.49577	
4/4-20-Q-1	1	0	31	0	0.50284	0.24561	0.74845	
4/4-20-Q-1	1	1	31	0	0.46336	0.22679	0.69015	
4/4-20-Q-1	1	2	28	0	0.39956	0.19612	0.59568	

Datewise verification for the first declared seed (continued)								
Service	Attempt	Date	Changed	Extra	$U - C$	$C - L$	$U - L$	
4/4-20-Q-1	1	3	15	0	0.63704	0.05329	0.69033	
4/4-20-Q-1	2	0	63	0	0.47865	0.20779	0.68644	
4/4-20-Q-1	2	1	63	0	0.43964	0.19181	0.63144	
4/4-20-Q-1	2	2	59	0	0.37687	0.16586	0.54273	
4/4-20-Q-1	2	3	42	0	0.63602	0.04785	0.68386	
8/6-20-Q-1	1	0	26	0	0.60302	0.41769	1.02070	
8/6-20-Q-1	1	1	24	0	0.58712	0.40623	0.99334	
8/6-20-Q-1	1	2	24	0	0.56153	0.38781	0.94934	
8/6-20-Q-1	1	3	20	0	0.52013	0.35796	0.87809	
8/6-20-Q-1	1	4	14	0	0.45268	0.30907	0.76175	
8/6-20-Q-1	1	5	0	0	0.63956	0.07309	0.71264	
8/6-20-Q-1	2	0	60	0	0.58520	0.37453	0.95973	
8/6-20-Q-1	2	1	60	0	0.56954	0.36417	0.93371	
8/6-20-Q-1	2	2	59	0	0.54452	0.34754	0.89205	
8/6-20-Q-1	2	3	54	0	0.50409	0.32066	0.82475	
8/6-20-Q-1	2	4	46	0	0.43849	0.27676	0.71525	
8/6-20-Q-1	2	5	0	0	0.63853	0.06811	0.70663	
2/2-20-X-1	1	0	30	0	0.31866	0.08581	0.40446	
2/2-20-X-1	1	1	27	0	0.60626	0.05340	0.65965	
2/2-20-X-1	2	0	62	1	0.26553	0.05781	0.31615	
2/2-20-X-1	2	1	58	0	0.45576	0.04002	0.49577	
4/4-20-X-1	1	0	31	0	0.50284	0.24561	0.74845	
4/4-20-X-1	1	1	31	0	0.46336	0.22679	0.69015	
4/4-20-X-1	1	2	28	0	0.39956	0.19612	0.59568	
4/4-20-X-1	1	3	15	0	0.63704	0.05329	0.69033	
4/4-20-X-1	2	0	63	0	0.47865	0.20779	0.68644	
4/4-20-X-1	2	1	63	0	0.43964	0.19181	0.63144	
4/4-20-X-1	2	2	59	0	0.37687	0.16586	0.54273	
4/4-20-X-1	2	3	42	0	0.63602	0.04785	0.68386	
8/6-20-X-1	1	0	26	0	0.60302	0.41769	1.02070	
8/6-20-X-1	1	1	24	0	0.58712	0.40623	0.99334	
8/6-20-X-1	1	2	24	0	0.56153	0.38781	0.94934	

Datewise verification for the first declared seed (continued)								
Service	Attempt	Date	Changed	Extra	$U - C$	$C - L$	$U - L$	
8/6-20-X-1	1	3	20	0	0.52013	0.35796	0.87809	
8/6-20-X-1	1	4	14	0	0.45268	0.30907	0.76175	
8/6-20-X-1	1	5	0	0	0.63956	0.07309	0.71264	
8/6-20-X-1	2	0	60	0	0.58520	0.37453	0.95973	
8/6-20-X-1	2	1	60	0	0.56954	0.36417	0.93371	
8/6-20-X-1	2	2	59	0	0.54452	0.34754	0.89205	
8/6-20-X-1	2	3	54	0	0.50409	0.32066	0.82475	
8/6-20-X-1	2	4	46	0	0.43849	0.27676	0.71525	
8/6-20-X-1	2	5	0	0	0.63853	0.06811	0.70663	
2/2-20-N-1	1	0	30	0	0.31866	0.08581	0.40446	
2/2-20-N-1	1	1	27	0	0.60626	0.05340	0.65965	
2/2-20-N-1	2	0	62	0	0.26553	0.05781	0.31615	
2/2-20-N-1	2	1	58	0	0.45576	0.04002	0.49577	
4/4-20-N-1	1	0	31	0	0.50284	0.24561	0.74845	
4/4-20-N-1	1	1	31	0	0.46336	0.22679	0.69015	
4/4-20-N-1	1	2	28	0	0.39956	0.19612	0.59568	
4/4-20-N-1	1	3	15	0	0.63704	0.05329	0.69033	
4/4-20-N-1	2	0	63	0	0.47865	0.20779	0.68644	
4/4-20-N-1	2	1	63	0	0.43964	0.19181	0.63144	
4/4-20-N-1	2	2	59	0	0.37687	0.16586	0.54273	
4/4-20-N-1	2	3	42	0	0.63602	0.04785	0.68386	
8/6-20-N-1	1	0	26	0	0.60302	0.41769	1.02070	
8/6-20-N-1	1	1	24	0	0.58712	0.40623	0.99334	
8/6-20-N-1	1	2	24	0	0.56153	0.38781	0.94934	
8/6-20-N-1	1	3	20	0	0.52013	0.35796	0.87809	
8/6-20-N-1	1	4	14	0	0.45268	0.30907	0.76175	
8/6-20-N-1	1	5	0	0	0.63956	0.07309	0.71264	
8/6-20-N-1	2	0	60	0	0.58520	0.37453	0.95973	
8/6-20-N-1	2	1	60	0	0.56954	0.36417	0.93371	
8/6-20-N-1	2	2	59	0	0.54452	0.34754	0.89205	
8/6-20-N-1	2	3	54	0	0.50409	0.32066	0.82475	
8/6-20-N-1	2	4	46	0	0.43849	0.27676	0.71525	

Datewise verification for the first declared seed (continued)							
Service	Attempt	Date	Changed	Extra	$U - C$	$C - L$	$U - L$
8/6-20-N-1	2	5	0	0	0.63853	0.06811	0.70663

This printed subset uses the first prespecified seed and the twenty-percent target for all modes. Every seed and target is in the machine-readable audit. Each numerical entry is a maximum over leaves; maxima of the first two components need not add to the maximum total gap. The displayed first-seed subset is specified mechanically, not chosen for favorable outcomes. Full datewise arrays for every seed, target and attempted policy remain in the machine-readable replay and raw records. For each leaf the identity $(U - L) = (U - C) + (C - L)$ is exact over the recorded endpoints; printed entries are separate outward-rounded maxima. Acquisition, innovation and arithmetic effects can contribute to both components. The analytic fitted-feature moduli in the article supply an additional primitive account and are not misreported as a separately executed six-factor numerical experiment.

The R57 article adds the fitted-witness transfer theorem, its action-sensitive neural moduli and the new prospective experiment to the same NBO manuscript. The full preceding article, supplement, response, broader applications and unfavorable comparisons remain in the preserved and complete development documents. The release checks the inherited label sets and scientific-source/result identities, compiles all seven active and companion documents and rebuilds them from a clean archive. Those checks establish reproducibility and internal consistency, not an external editorial decision.

7. LOSSLESS NEURAL-SEARCH RECORDS

The new main-article results concern exact action recovery and the invariance of a mathematical prospective controller. This supplement identifies the finite computations used to evaluate them. The frozen R58 scientific code and protocol are preserved without modification; R59 adds the proof, manuscript integration and complete record-reconstruction audit. The audit does not retrain models or replace their measured service clocks.

7.1. *Feature reduction, arithmetic and fallback*

The stored ReLU coefficients and acquired-cell centers are interpreted as exact binary rationals. Action-dependent projections and activation-region tests are evaluated rationally. A ridge wholly inside one region on the full robust feasible interval is incorporated into the polynomial part, with its constant retained separately. Crossing ridges, including those with negative output weights, are evaluated by outward interval arithmetic. The expectation formula for a zero innovation projection is the ordinary positive part; division by a zero radius is not attempted.

The endpoint calculation evaluates every lattice index with outward conversion, arithmetic and pairwise summation. It does not use an unrounded matrix product as a rigorous endpoint. A point survives when its lower endpoint is no larger than the smallest upper endpoint. Exact rational evaluation of the original fitted objective, rather than its reduced floating-point midpoint, chooses the return and resolves ties. The implementation checks containment of the returned exact value in its interval.

The screening limit is 2049 lattice points. A larger lattice, nonfinite arithmetic or an invalid interval triggers the inherited exact algebraic solver. This fallback preserves exactness but can cost more than a successful screen; its occurrence is explicitly counted. The present production catalogue has no fallback. Small disjoint correctness fixtures exercise both fallback paths, exact ties, activation boundaries, signed weights, zero capacity, zero innovation projection, non-dyadic action spacing and exhaustive lattice agreement. These tests are not additional economic observations.

TABLE X: Exact activation-region reduction of trained ridges

(d, T)	Reduction	Crossing ridges	Reduced ridges	Survivors	Fallbacks
(2,2)	10%	37	987	32	0
(2,2)	20%	136	2936	96	0
(4,4)	10%	178	2894	96	0
(4,4)	20%	599	8617	288	0
(8,6)	10%	317	4803	160	0
(8,6)	20%	1054	14306	480	0

Counts aggregate ridge occurrences across neural queries and attempted resolutions, not the number of distinct trained features. Reduced ridges are incorporated exactly into polynomial coefficients or vanish identically. Their weights are not thresholded.

The survivor-gap and quadratic-growth statements are conditional analytical results. The production audit records actual survivors and endpoint hashes. It does not infer a global growth constant from those observed survivor counts. Similarly, the interval arithmetic count is distinct from a bit-complexity bound: exact rational operation cost depends on coefficient sizes, which are not assumed to be one machine word.

8. EVERY TIMING REPETITION AND PAID WORK COMPONENT

TABLE XI: Every complete-process timing repetition

(d, T)	Reduction	Mode	Rep. 1 (s)	Rep. 2 (s)	Rep. 3 (s)	Median (s)
(2,2)	10%	A	1.234	1.212	1.173	1.212
(2,2)	10%	S	1.097	1.101	1.147	1.101
(2,2)	20%	A	3.577	3.775	3.565	3.577
(2,2)	20%	S	3.374	3.409	3.640	3.409
(4,4)	10%	A	5.973	5.837	5.808	5.837
(4,4)	10%	S	5.487	5.494	5.456	5.487
(4,4)	20%	A	20.646	20.554	20.489	20.554
(4,4)	20%	S	19.373	19.396	19.344	19.373
(8,6)	10%	A	20.239	20.229	20.213	20.229
(8,6)	10%	S	19.785	19.223	19.270	19.270
(8,6)	20%	A	63.827	64.581	63.814	63.827
(8,6)	20%	S	61.842	62.079	62.075	62.075

No timing trial is selected or removed. A is algebraic and S screened. Fits and inference streams deliberately repeat; these are timing repetitions, not independent learned objects or additional policy-cost samples.

TABLE XII: Recorded service components and surrounding-process remainder

(d, T)	Reduction	Mode	Fit (s)	Search (s)	Verify (s)	Inference (s)	Other (s)
(2,2)	10%	A	0.122	0.178	0.367	0.146	0.384
(2,2)	10%	S	0.121	0.127	0.366	0.146	0.341
(2,2)	20%	A	0.466	0.495	1.043	1.188	0.381
(2,2)	20%	S	0.466	0.321	1.042	1.216	0.367
(4,4)	10%	A	0.492	0.878	3.907	0.127	0.435
(4,4)	10%	S	0.493	0.507	3.919	0.127	0.431
(4,4)	20%	A	1.891	2.530	11.607	3.999	0.492
(4,4)	20%	S	1.890	1.365	11.631	4.014	0.488
(8,6)	10%	A	1.057	1.626	15.742	1.247	0.536
(8,6)	10%	S	1.046	0.989	15.490	1.252	0.492
(8,6)	20%	A	4.398	4.901	47.688	6.243	0.638
(8,6)	20%	S	4.463	2.760	47.925	6.255	0.611

Entries are componentwise medians across the three services. Their sum need not equal the median complete-process clock. Other is calculated separately within each service as its surrounding clock minus the four component clocks; it includes imports, record handling, output and termination, not parent-controller checkpoint overhead. Search includes the common terminal solver.

TABLE XIII: Complete 36-service return, work and storage registry

(d, T)	Reduction	Mode	Rep.	Return	Attempts	Time (s)	MiB out	MiB RSS
(2,2)	20%	A	1	M	2	3.577	5.46	127.42
(2,2)	20%	A	2	M	2	3.775	5.46	127.77
(2,2)	20%	A	3	M	2	3.565	5.46	128.40
(2,2)	10%	A	1	M	1	1.234	0.73	80.54
(2,2)	10%	A	2	M	1	1.212	0.73	80.50
(2,2)	10%	A	3	M	1	1.173	0.73	80.50
(4,4)	20%	A	1	M	2	20.646	6.17	130.63
(4,4)	20%	A	2	M	2	20.554	6.17	130.07
(4,4)	20%	A	3	M	2	20.489	6.17	130.85
(4,4)	10%	A	1	M	1	5.973	0.43	80.56
(4,4)	10%	A	2	M	1	5.837	0.43	80.50
(4,4)	10%	A	3	M	1	5.808	0.43	80.69

	(d, T)	Reduction	Mode	Rep.	Return	Attempts	Time (s)	MiB out	MiB RSS	
1										1
2	(8,6)	20%	A	1	E	2	63.827	5.82	132.68	2
3	(8,6)	20%	A	2	E	2	64.581	5.82	133.25	3
4	(8,6)	20%	A	3	E	2	63.814	5.82	132.90	4
5	(8,6)	10%	A	1	M	1	20.239	1.26	81.27	5
6	(8,6)	10%	A	2	M	1	20.229	1.26	81.43	6
7	(8,6)	10%	A	3	M	1	20.213	1.26	81.48	7
8	(2,2)	20%	S	1	M	2	3.374	5.49	128.04	8
9	(2,2)	20%	S	2	M	2	3.409	5.49	127.97	9
10	(2,2)	20%	S	3	M	2	3.640	5.49	128.73	10
11	(2,2)	10%	S	1	M	1	1.097	0.74	80.70	11
12	(2,2)	10%	S	2	M	1	1.101	0.74	80.56	12
13	(2,2)	10%	S	3	M	1	1.147	0.74	80.59	13
14	(4,4)	20%	S	1	M	2	19.373	6.27	128.40	14
15	(4,4)	20%	S	2	M	2	19.396	6.27	128.80	15
16	(4,4)	20%	S	3	M	2	19.344	6.27	129.39	16
17	(4,4)	10%	S	1	M	1	5.487	0.46	80.42	17
18	(4,4)	10%	S	2	M	1	5.494	0.46	80.48	18
19	(4,4)	10%	S	3	M	1	5.456	0.46	80.62	19
20	(8,6)	20%	S	1	E	2	61.842	6.00	130.35	20
21	(8,6)	20%	S	2	E	2	62.079	6.00	130.79	21
22	(8,6)	20%	S	3	E	2	62.075	6.00	130.88	22
23	(8,6)	10%	S	1	M	1	19.785	1.32	81.41	23
24	(8,6)	10%	S	2	M	1	19.223	1.32	81.63	24
25	(8,6)	10%	S	3	M	1	19.270	1.32	81.32	25

M means target attained and E budget exhausted. Output bytes are complete serialized own-service files; RSS is process peak resident memory. These are not the logical working storage of the screening kernel. Exact service keys, parameter and policy identities, stagewise endpoints, raw arrays and clocks remain in the machine-readable audit. For each task, mode order alternates across repetition and target, processes run sequentially with fixed available-core affinity, and numerical-library thread counts are one. The three repetitions have the same fitted parameters and interval-path streams; they measure repeated work rather than independent training or inference variation. Governor availability and environment versions are

recorded in each service JSON. Frequency is not controlled, and different tasks use different runners.

The surrounding clock ends after process termination and log flush. It contains source-freeze verification, imports, fitting, all candidate search, verification, interval-path evaluation, inference and serialized service records. The controller then writes its checkpoint containing that elapsed value. This checkpoint operation is outside the surrounding service clock. Accordingly the table is complete for the measured service process and log, not a claim that the surrounding orchestration has no cost.

The component medians are descriptive. They need not sum to the median whole-process time. Output bytes refer to all files of the individual service, including intermediate attempts and raw arrays; process peak resident memory includes interpreter and numerical-library state. Neither replaces the vectorized screen’s logical working-storage bound.

9. DATEWISE ACCURACY AND THE INVARIANT STOPPING HISTORY

TABLE XIV: Datewise decisions and unchanged full-action certificate components

(d, T)	Reduction	Stage	Date	Changed	Extra	$U - L$	$U - C$	$C - L$
(2,2)	20%	0	0	30	0	0.40446	0.31866	0.08581
(2,2)	20%	0	1	27	0	0.65965	0.60626	0.05340
(2,2)	20%	1	0	62	1	0.31615	0.26553	0.05781
(2,2)	20%	1	1	58	0	0.49577	0.45576	0.04002
(2,2)	10%	0	0	30	0	0.40446	0.31866	0.08581
(2,2)	10%	0	1	27	0	0.65965	0.60626	0.05340
(4,4)	20%	0	0	31	0	0.74845	0.50284	0.24561
(4,4)	20%	0	1	31	0	0.69015	0.46336	0.22679
(4,4)	20%	0	2	28	0	0.59568	0.39956	0.19612
(4,4)	20%	0	3	15	0	0.69033	0.63704	0.05329
(4,4)	20%	1	0	63	0	0.68644	0.47865	0.20779
(4,4)	20%	1	1	63	0	0.63144	0.43964	0.19181
(4,4)	20%	1	2	59	0	0.54273	0.37687	0.16586
(4,4)	20%	1	3	42	0	0.68386	0.63602	0.04785
(4,4)	10%	0	0	31	0	0.74845	0.50284	0.24561

	(d, T)	Reduction	Stage	Date	Changed	Extra	$U - L$	$U - C$	$C - L$	
1										1
2	(4,4)	10%	0	1	31	0	0.69015	0.46336	0.22679	2
3	(4,4)	10%	0	2	28	0	0.59568	0.39956	0.19612	3
4	(4,4)	10%	0	3	15	0	0.69033	0.63704	0.05329	4
5	(8,6)	20%	0	0	26	0	1.02070	0.60302	0.41769	5
6	(8,6)	20%	0	1	24	0	0.99334	0.58712	0.40623	6
7	(8,6)	20%	0	2	24	0	0.94934	0.56153	0.38781	7
8	(8,6)	20%	0	3	20	0	0.87809	0.52013	0.35796	8
9	(8,6)	20%	0	4	14	0	0.76175	0.45268	0.30907	9
10	(8,6)	20%	0	5	0	0	0.71264	0.63956	0.07309	10
11	(8,6)	20%	1	0	60	0	0.95973	0.58520	0.37453	11
12	(8,6)	20%	1	1	60	0	0.93371	0.56954	0.36417	12
13	(8,6)	20%	1	2	59	0	0.89205	0.54452	0.34754	13
14	(8,6)	20%	1	3	54	0	0.82475	0.50409	0.32066	14
15	(8,6)	20%	1	4	46	0	0.71525	0.43849	0.27676	15
16	(8,6)	20%	1	5	0	0	0.70663	0.63853	0.06811	16
17	(8,6)	10%	0	0	26	0	1.02070	0.60302	0.41769	17
18	(8,6)	10%	0	1	24	0	0.99334	0.58712	0.40623	18
19	(8,6)	10%	0	2	24	0	0.94934	0.56153	0.38781	19
20	(8,6)	10%	0	3	20	0	0.87809	0.52013	0.35796	20
21	(8,6)	10%	0	4	14	0	0.76175	0.45268	0.30907	21
22	(8,6)	10%	0	5	0	0	0.71264	0.63956	0.07309	22

One of the exactly matching solver/repetition transcripts is printed to avoid duplicating numerical identities. Extra means a selected action changed by adding the exact fitted witness to the common plus critic-menu candidates. Each bound is rounded upward. The component maxima need not sum to the maximum total gap, and the local gap is not substituted for final Bellman loss. Every record retains the common candidate menu, the learned menu proposal and exact additional witness, together with the whole-cell candidate endpoints and continuous-action lower covers. The raw fields also retain the selected upper bound U , full-action lower bound L , intermediary C , selected action and the policy obtained without the additional exact witness. The identity

$$U - L = (U - C) + (C - L)$$

holds cellwise; maxima taken separately need not add to the maximum total. The extra-witness column identifies an actual change in the selected action, not merely a smaller fitted objective at the center.

These datewise quantities are local greedy-error certificates. The returned policy's all-state loss account also includes a valid bound for the installed reference continuation and is capped by the original total-cost support. The audit reconstructs that exact rational expression from every saved stage. Exact fitted optimization does not remove the full-action, acquisition, continuation or verification terms. The initial-law cost target and original Bellman-loss objective remain separate contracts.

For every saved inference look, the audit regenerates the declared bins, evaluates both the returned policy and installed zero policy on the corresponding continuous-bin intervals, and compares every lower and upper path endpoint. It verifies cost, gain and target-contrast bounds using the inherited exact-rational moment checks. The target classification is reconstructed from the exact endpoint, not from a rounded printed interval. A successful or positive-lower look forbids another look at that stage; success also forbids another construction attempt. Final exhaustion requires all declared attempts to have been used.

The reconstructed R58 catalogue has nine distinct task–target–attempt path streams. It has six distinct fitted-model verification inputs; both exact solvers are evaluated on each. All repeated instances still undergo file, parameter, policy, certificate and path-archive checks. Memoization avoids calling identical deterministic calculations independent observations. It does not omit an adverse service or permit an unmatched input to reuse another input's proof.

10. SOURCE PROVENANCE AND THE SCOPE OF THE REVISION

The controlling review is the 9 October 2026 advisory report on R54. The complete R57 manuscript supplies the original constructive operator, trained-neural transfer theory, non-tensor prospective method and conventional comparisons. The later R58 source-frozen execution supplies the matched exact-search experiment. R59 starts from the pinned R58 development commit and retains its entire scientific source and evidence history in the new revision directory.

The paper, supplement and response are ordinary sources; reproduction does not assemble mathematical claims from expiring downloads. The release checks inherited source/result hashes and labeled results, runs the previous test and audit chain, reconstructs the new matched catalogue, compiles the seven-document family and rebuilds it from a clean archive after removing generated outputs. The release manifest identifies the tested source and final remote package. These checks establish traceable execution and publication consistency, not external mathematical or editorial acceptance.

The economic observations retain their original cohort labels and confidence families. R59 publication adds zero training services and zero independent policy-cost samples. The equal-policy identity is exact, whereas the separate neural-versus-conventional comparisons retain their observed interval classifications. The broader NBO applications remain available under their own hypotheses and are not silently assigned the expected-cost theorem's assumptions.

REFERENCES